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UNIVERSITY OF TORONTO
Faculty of Arts and Science

APRIL 2018 EXAMINATIONS

CSC 336 H1S
Instructor: Fairgrieve

Duration — 3 hours

Examination Aids: **One $8\frac{1}{2}'' \times 11''$ sheet of paper, handwritten on both sides.**
No photocopies.
A calculator.

You must earn at least 27 out of 90 marks (30%) on this final examination in order to pass the course. Otherwise, your final course grade will be no higher than 47%.

Student Number: |_|_|_|_|_|_|_|_|_|_|_|

Family Name(s): _____

Given Name(s): _____

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*Do **not** turn this page until you have received the signal to start.*

This final examination paper consists of 12 questions on 16 pages (including this one). When you receive the signal to start, please make sure that your copy of the final examination is complete.

Answer each question directly on this paper, in the space provided. Blank pages can be found at the end of the exam and may be used for rough work or in case you need more space for solutions.

THIS EXAM IS DOUBLE-SIDED.

1: _____/ 6
2: _____/ 8
3: _____/ 6
4: _____/ 6
5: _____/10
6: _____/ 9
7: _____/ 7
8: _____/ 8
9: _____/ 8
10: _____/ 7
11: _____/ 8
12: _____/ 7

TOTAL: _____/90

Good Luck!

CONT'D...

Question 1. [6 MARKS]

We have observed that we should use numerically stable algorithms when we need to compute accurate solutions to mathematical problems. Name **two** algorithms that were discussed in the course that you know to be numerically **unstable** but which can be modified to give a numerically stable algorithm. For each algorithm, state the mathematical problem the original algorithm is meant to solve, the source of the numerical instability, and briefly explain how the algorithm can be modified to produce a numerically **stable** algorithm.

Part (a) [3 MARKS]

Algorithm 1:

Part (b) [3 MARKS]

Algorithm 2:

Question 2. [8 MARKS]

Part (a) [2 MARKS]

List two reasons why **binary** floating-point number systems are usually normalized.

Part (b) [2 MARKS]

What properties of a floating-point number system determine the maximum relative error when representing a given real number by a floating-point number?

Part (c) [1 MARK]

In floating-point arithmetic, which of the following operations on two **positive** floating-point operands can produce an overflow?

`(a) Addition` `(b) Subtraction` `(c) Multiplication` `(d) Division`

Part (d) [2 MARKS]

For what values of x might the straightforward evaluation of the expression $\frac{1}{1+2x} - \frac{1-x}{1+x}$ suffer from catastrophic cancellation?

Part (e) [1 MARK]

Rewrite the expression $\frac{1}{1+2x} - \frac{1-x}{1+x}$ to get an equivalent form that avoids catastrophic cancellation.

Question 3. [6 MARKS]

The Python programming language `math` module contains two functions for evaluating logarithms to the base e, namely `math.log()` and `math.log1p()`. They both have a type `float` parameter and return a type `float` value.

The `math.log(x)` function computes an approximation to the natural logarithm of `x`, $\log_e(x)$. The value of `x` must be positive. The `math.log1p(y)` function computes an approximation to $\log_e(1.0 + y)$. The value of `y` must be greater than -1.0.

A student who has not yet taken CSC 336 might believe that it is redundant to include two such logarithm functions in the `math` module. Explain why the `math` module contains both the `log()` and `log1p()` functions.

Question 4. [6 MARKS]

Part (a) [2 MARKS]

Suppose $\underline{\hat{x}}$ is a computed solution to the linear system $\mathbf{A}\underline{x} = \underline{b}$ and that

$$\|\underline{\hat{x}}\|_\infty \approx 2.5, \quad \|\underline{b}\|_\infty \approx 7.3, \quad \|\underline{b} - \mathbf{A}\underline{\hat{x}}\|_\infty \approx 2.4 \times 10^{-14}, \quad \|\mathbf{A}\|_\infty \approx 1.3 \times 10^2 \quad \text{and} \quad \|\mathbf{A}^{-1}\|_\infty \approx 4.0 \times 10^4.$$

Assume that $\underline{\hat{x}}$ has enough accuracy to make it reasonable to approximate $\|\underline{x}^*\|_\infty$ by $\|\underline{\hat{x}}\|_\infty$. The vector $\underline{x}^*$ is the true solution to the problem.

Determine an upper bound for the absolute error in each component of the computed solution $\underline{\hat{x}}$.

Part (b) [4 MARKS]

It is known that the components of the exact solution to a linear system $\mathbf{A}\underline{x} = \underline{b}$ range in size from 10^{-1} to 10^3 and that the ∞-norm condition number of $\mathbf{A}$ is about 10^4. How many significant digits must your computer use in order to ensure that the **smallest** component in the solution computed using Gaussian Elimination with Partial Pivoting has at least five digits of accuracy? A rigorous proof is **not** required, just a reasonable heuristic argument. State any assumptions or approximations used.

Question 5. [10 MARKS]

Part (a) [8 MARKS]

When a numerical linear algebra algorithm is able to perform its work without the creation of additional array storage, it is said to operate *in place*.

Write an algorithm for computing the inverse of a **nonsingular** $n \times n$ **lower-triangular** matrix named $\mathbf{A}$ that operates *in place*. The creation of temporary scalar variables is allowed. But do not create any matrices or vectors. Assume that pivoting is not necessary.

HINT: The inverse of a nonsingular lower-triangular matrix is also lower-triangular, and so your algorithm only needs to change elements in the lower-triangular part of $\mathbf{A}$.

MORE HINT: Start by thinking about what mathematical problem needs to be solved, and then write an algorithm to do it.

Part (b) [2 MARKS]

Determine the operation count for the algorithm developed in Part (a), measured in *flops*. Show your work.

Question 6. [9 MARKS]

Let $f(x) = x^2 - x - 0.24$ and $g(x) = x^2 - 0.24$. A value x^* is a root of $f(x)$ if and only if it is a fixed-point of $g(x)$.

Part (a) [1 MARK]

Show that $P_1 = -0.2$ is a fixed-point of $g(x)$.

Part (b) [4 MARKS]

Consider the fixed-point iteration $x_{i+1} = g(x_i)$, $i = 0, 1, 2, \ldots$, for some given x_0 that is not too far from P_1. Do you expect that the fixed-point iteration would calculate the fixed-point at -0.2 to 10 correct decimal places faster or slower than the Bisection Method? Justify your conclusion.

Part (c) [2 MARKS]

Determine the other fixed-point of $g(x)$. Label this value P_2.

Part (d) [2 MARKS]

Will the fixed-point iteration $x_{i+1} = g(x_i)$, $i = 0, 1, 2, \ldots$, converge to the second fixed-point (P_2) if $x_0 \neq P_2$? Justify your conclusion.

Question 7. [7 MARKS]

Newton's method is sometimes used to implement the built-in square root function on a computer. The initial guess can be taken from a table of approximate solutions.

Part (a) [4 MARKS]

What is the Newton's method iteration for computing the square root of a positive number y (i.e. for solving the equation $f(x) = x^2 - y = 0$ for x, given y)? Simplify your expression to reduce the number of floating-point operations.

Part (b) [3 MARKS]

If we assume that the starting guess has an accuracy of at least 4 bits, how many iterations would you expect to be necessary to obtain a result with:

1. 24 bits of accuracy? Explain.

2. 53 bits of accuracy? Explain.

Question 8. [8 MARKS]

Part (a) [4 MARKS]

If the errors at successive iterations of an iterative method are as follows, how would you characterize the convergence rates? Explain your answer.

Errors: $10^{-2}, 10^{-4}, 10^{-8}, 10^{-16}, \ldots$

Errors: $10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, \ldots$

Part (b) [4 MARKS]

The values of `x(i), i=0,1,2,3,...,` in the following table of computational results appear to be converging to `6.0`. Estimate the rate of convergence. Explain your answer.

```
i   x(i)                    | x(i)-x(i-1) |
==  ====================    ===============
 0  6.04057907766532e+00
 1  6.00910642088410e+00    3.15e-02
 2  6.00140654830877e+00    7.70e-03
 3  6.00013619567625e+00    1.27e-03
 4  6.00000735654939e+00    1.29e-04
 5  6.00000019156347e+00    7.16e-06
 6  6.00000000200383e+00    1.90e-07
 7  6.00000000000670e+00    2.00e-09
 8  6.00000000000001e+00    6.70e-12
 9  6.00000000000000e+00    5.33e-15
```

Question 9. [8 MARKS]

Part (a) [2 MARKS]

List one advantage and one disadvantage of Lagrange interpolation compared with using the monomial basis for polynomial interpolation.

Part (b) [2 MARKS]

When fitting a large number of data points, what is the main advantage of piecewise polynomial interpolation over interpolation by a single polynomial?

Part (c) [2 MARKS]

How many times is a piecewise cubic Hermite interpolant guaranteed to be continuously differentiable? (i.e., how many continuous derivatives must a piecewise cubic Hermite interpolant have?)

answer:

How many times is a cubic spline interpolant guaranteed to be continuously differentiable? (i.e., how many continuous derivatives must a cubic spline interpolant have?)

answer:

Part (d) [2 MARKS]

Is it ever possible for two distinct polynomials to interpolate the same m data points? If so, under what conditions, and if not, why?

Question 10. [7 MARKS]

Part (a) [3 MARKS]

Given the three data points

x	-1	0	1
y	-1	0	1

,

use a **Newton basis** to construct the polynomial $p_2(x)$ of degree at most 2 that interpolates the data. Show your calculations.

Part (b) [4 MARKS]

The data given in part (a) was generated by evaluating the function $f(x) = \sin(\frac{\pi}{2}x)$ at the points $x = -1$, 0 and 1. Use the error bound formula for polynomial interpolation to derive an accurate upper bound for

$$\max_{x \in [-1,1]} |f(x) - p_2(x)|.$$

Give a numerical value for your bound. Show your calculations.

Question 11. [8 MARKS]

On Assignment Four you used the inverse interpolation technique to estimate the percentiles of the error function $\mathrm{erf}(x)$. You used a cubic spline to construct the inverse interpolant, but for this question, you are to instead use a single polynomial to construct the inverse interpolant.

Use the inverse interpolation technique to estimate a value of x such that $f(x) = 0$, given the following table of values for f :

x	-2	0	1
$f(x)$	-7	1	2

Question 12. [7 MARKS]

Part (a) [5 MARKS]

Determine a, b and c so that $S(x)$ is a cubic spline function, where

$$S(x) = \begin{cases} x^3, & x \in [0,1], \\ -\frac{1}{2}(x-1)^3 + a(x-1)^2 + b(x-1) + c, & x \in [1,3]. \end{cases}$$

Part (b) [2 MARKS]

Is $S(x)$ a **natural** cubic spline?

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Total Marks = 90